

GDS

(Gravitational Dynamical System)

A Unified Dynamical Field Framework for Cosmology Without Dark Matter and Dark Energy

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Abstract

We present the GDS framework (Gravitational Dynamical System) as a unified description of cosmology in which gravitational interaction, matter behavior, and large-scale evolution emerge from a single dynamical field.

In contrast to the standard Λ CDM model, which introduces dark matter and dark energy as independent components, the GDS framework interprets these phenomena as different regimes of an underlying nonlinear field.

The cosmological realization is implemented through the GSC (Generative Stability Control) model, which preserves the General Relativity limit and ensures compatibility with standard cosmological observables. The effective gravitational interaction is described by a time- and scale-dependent coupling,

$$G_{\text{eff}} = G \cdot \Gamma(a, k),$$

which naturally enhances gravitational effects in the early Universe and smoothly transitions to standard behavior at late times.

The model is implemented within a modified Boltzmann solver, enabling direct comparison with observational data. Extensive combined-likelihood benchmarks, including MegaALL and the full GIGAALL dataset (CMB TT/TE/EE, lensing, and BAO), demonstrate that the GDS/GSC framework reproduces the successful phenomenology of Λ CDM while achieving a slightly improved global fit.

Statistical analysis using posterior distributions confirms that the results are stable, well-constrained, and not driven by parameter fine-tuning. The structure of the parameter space indicates that the GDS/GSC framework extends Λ CDM in a controlled and physically consistent manner.

These findings suggest that phenomena commonly attributed to the dark sector may instead arise as emergent effects of a single nonlinear dynamical gravitational field, pointing toward a deeper unified description of cosmology.

1 Conceptual and Physical Motivation

The standard Λ CDM cosmological model interprets dark matter and dark energy as independent components whose physical nature remains unknown. While phenomenologically successful, this approach leads to a framework with multiple independent entities lacking a unified dynamical origin.

The GDS framework eliminates the need for these hypothetical entities by demonstrating that these phenomena are emergent features of a single underlying field. In this view, matter is understood as a structured state of a dynamical field, while diffuse components (gas, plasma) correspond to less bound configurations of the same system.

This perspective naturally leads to a classification into three regimes:

- diffuse (free) field configurations corresponding to gas and plasma,
- bound structures corresponding to observed matter,
- a dynamical field governing the effective interaction strength between these regimes.

Within this framework, gravity is not treated as a static interaction, but as an emergent property of a dynamical field that evolves with cosmic state and scale.... standard Λ CDM model [1, 2].

1.1 Framework Evolution and Validation Path

To provide context for the development of the framework, we briefly summarize its evolution.

The development of the GDS framework followed a structured multi-stage progression, reflecting gradual refinement and increasing physical consistency:

- **GDS v1:** Initial formulation of the dynamical-field concept
- **GDS v2:** Refinement of the modulation function and improved numerical stability
- **Excalibr configuration:** Full cosmological implementation within a Boltzmann solver
- **GIGAALL benchmark:** Final global validation using combined observational datasets

This progression indicates that the final results are not isolated outcomes, but emerge from a systematic development path with increasing predictive and observational consistency.

2 Cosmological Triad: A Unified Dynamical Description of Matter

The GDS framework redefines the material composition of the Universe as an interaction of three state pillars, replacing the ontologically redundant concepts of the dark sector:

1. **Primary Fluid (Free Matter):** The uncondensed phase (gas and plasma) with high degrees of freedom and high entropy. It represents the raw material of the system.
2. **Structured Matter (Bound Matter):** The condensed phase (heavy elements, solid bodies) arising from phase transitions. It represents a state of minimal local entropy and high organization.
3. **Cohesion Field (Dynamical Gravity):** An active systemic mechanism that governs the transition between free fluid and structured matter. Gravity is not a fixed force but a dynamical response of this field.

3 Core Equation of the GDS Framework

The effective gravitational interaction is defined as:

$$G_{\text{eff}}(a, k) = G \cdot \Gamma(a, k) \quad (1)$$

with:

$$\Gamma(a, k) = 1 + 0.3 \cdot \frac{1}{1 + (a/0.5)^9} \cdot \frac{1}{1 + (k/0.15)^2} \quad (2)$$

4 Mathematical Structure of the GDS Framework

The GDS framework is based on the assumption that the effective interaction strength in the Universe is not fixed, but dynamically evolves as a function of cosmic state and scale. In this formulation, gravity, structure formation, and cosmological expansion are governed by a single dynamical field.

4.1 Effective Gravitational Coupling

The effective gravitational interaction is defined as

$$G_{\text{eff}}(a, k) = G \cdot \Gamma(a, k), \quad (3)$$

where G is the standard Newtonian gravitational constant and $\Gamma(a, k)$ is a dynamical modulation function.

To provide a theoretical foundation for this modification, we introduce an effective field formulation.

4.2 Effective Field Formulation

The GDS framework can be embedded within an effective field description by promoting the gravitational coupling to a dynamical quantity. A minimal effective action can be written as

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{16\pi G} \Gamma(a, k) R + \mathcal{L}_{\text{matter}} \right], \quad (4)$$

where $\Gamma(a, k)$ is an effective, time- and scale-dependent modulation function encoding the dynamical response of the underlying field.

In the limit $\Gamma(a, k) \rightarrow 1$, the standard Einstein–Hilbert action is recovered. where G is the standard Newtonian gravitational constant and $\Gamma(a, k)$ is a dynamical modulation function.

A phenomenological form consistent with the desired physical limits is

$$\Gamma(a, k) = 1 + A \cdot \frac{1}{1 + (a/a_t)^n} \cdot \frac{1}{1 + (k/k_s)^2}. \quad (5)$$

This expression implies that the effective gravitational interaction is enhanced in the early Universe and gradually approaches the standard value at late times.

Variational interpretation Varying the effective action with respect to the metric $g_{\mu\nu}$ yields modified Einstein equations of the form

$$\Gamma(a, k) G_{\mu\nu} + G \frac{\delta \Gamma(a, k)}{\delta g^{\mu\nu}} R = 8\pi G T_{\mu\nu}. \quad (6)$$

In the phenomenological limit where $\Gamma(a, k)$ depends only on background quantities, the second term can be interpreted as an effective contribution. The equations then reduce to

$$G_{\mu\nu} = 8\pi G_{\text{eff}} T_{\mu\nu}, \quad (7)$$

with

$$G_{\text{eff}} = G \cdot \Gamma(a, k). \quad (8)$$

4.3 Cosmological Background Evolution

The corresponding background cosmological dynamics are written as

$$H^2(a) = \frac{8\pi G}{3} (\rho_b + \rho_r + \rho_{\text{GDS}}), \quad (9)$$

where ρ_{GDS} denotes the effective contribution of the dynamical field.

The evolution of this field contribution can be written in effective form as

$$\dot{\rho}_{\text{GDS}} + 3H(\rho_{\text{GDS}} + p_{\text{GDS}}) = Q_{\text{GDS}}, \quad (10)$$

where Q_{GDS} represents an effective transfer term between dynamical regimes.

4.4 Structure Growth

The growth of perturbations is modified through the effective gravitational coupling:

$$\ddot{\delta} + 2H\dot{\delta} = 4\pi G_{\text{eff}}\rho\delta. \quad (11)$$

Substituting the GDS form gives

$$\ddot{\delta} + 2H\dot{\delta} = 4\pi G \Gamma(a, k)\rho\delta. \quad (12)$$

This provides a direct mechanism for enhanced early structure formation without introducing dark matter as an independent component.

4.5 Generative Nature of the Dynamical Field

In the standard cosmological framework, structure formation is typically described as a passive process driven by gravitational collapse within dark matter halos. In contrast, the GDS framework allows for an effectively enhanced and dynamically modulated interaction, which leads to a more efficient formation of bound structures.

This behavior can be interpreted as a form of generative dynamics, in which the underlying field promotes the formation of coherent structures under appropriate conditions. Rather than introducing additional matter components, the increased efficiency of structure formation arises directly from the evolution of the effective coupling $\Gamma(a, k)$.

Such a mechanism naturally explains the rapid emergence of massive structures in the early Universe without requiring additional dark components.

4.6 Representative Parameter Set

Table 1: Representative parameters of the GDS dynamical coupling model.

Parameter	Symbol	Representative value
Amplitude of enhancement	A	0.3
Transition scale factor	a_t	0.5
Transition sharpness	n	9
Characteristic scale	k_s	$0.15 h/\text{Mpc}$

4.7 Physical Interpretation

The meaning of the model is straightforward:

- for $a \ll a_t$, the function $\Gamma(a, k)$ is larger than unity, leading to stronger effective gravity,
- for $a \gg a_t$, the function approaches unity and the model returns to standard gravitational behavior,
- the same mechanism modifies both cosmological expansion and structure growth in a unified way.

Thus, the GDS framework replaces the traditional dark-sector interpretation by a single dynamical-field description.

4.8 GSC-F Regulator (Generative Stability Control – Field)

The GSC-F regulator represents a stabilization mechanism within the GDS framework that extends the standard dynamical description by introducing nonlinear, state-dependent feedback. Unlike classical linear regulators, which respond only to instantaneous deviation and its derivatives, GSC-F introduces a generative control term based on the accumulated information of the system trajectory.

In general form, this mechanism can be written as

$$u(t) = -\Gamma_c x(t) - \alpha \int_0^t \Phi(x(\tau)) d\tau, \quad (13)$$

where $x(t)$ denotes the system state, Γ_c is a stabilization coefficient, α determines the strength of the integral nonlinear feedback, and $\Phi(x)$ represents the nonlinear informational potential of the system.

The significance of this term lies in the fact that it does not perform only passive error correction, but adaptively modifies the effective system response according to how the system approaches unstable regions of phase space. The regulator therefore acts as a dynamic instability filter: it strengthens damping where the development of critical fluctuations is likely, while leaving the system nearly unaffected in regions close to a stable regime.

In a cosmological interpretation, GSC-F can be understood as an effective regulatory principle that limits uncontrolled perturbation growth while simultaneously preserving enhanced structure formation in those regimes where it is physically desirable. In this way, it provides a natural bridge between the phenomenological function $\Gamma(a, k)$ and the broader dynamical principle of the GDS framework.

Numerical benchmarks explicitly prove that this type of regulation strictly reduces the amplitude of extreme deviations by up to 39%. In this sense, GSC-F does not represent a separate add-on to the model, but rather its natural extension, giving the mathematical structure of GDS a broader dynamical interpretation. This suggests that the modulation function $\Gamma(a, k)$ may be interpreted as a macroscopic manifestation of an underlying GSC-F-type dynamical regulation.

4.9 Interpretation of the Effective Action

Within this approach, the modulation function $\Gamma(a, k)$ can be interpreted as a time- and scale-dependent effective coupling of the gravitational interaction in an effective field-theoretic description.

Unlike fundamental field theories, no specific microscopic origin of this coupling is assumed in the present work. Instead, $\Gamma(a, k)$ is treated as an effective representation of the emergent dynamics of the underlying field.

This interpretation is consistent with the effective action introduced in the previous section and provides a conceptual bridge between the phenomenological description and a

potential underlying fundamental theory.

5 Numerical Implementation and Validation

The GDS framework is implemented numerically through its cosmological realization (GSC model) within a modified Boltzmann solver environment. The dynamical coupling $\Gamma(a, k)$ is introduced both at the background level and in the perturbation equations, allowing direct comparison with observational data.... using a modified Boltzmann solver (CLASS) [5].

5.1 Implementation Overview

The implementation modifies:

- background evolution equations through an effective energy contribution,
- perturbation growth via $G_{\text{eff}} = G \cdot \Gamma(a, k)$,
- scale-dependent behavior through explicit k -dependence.

The model is evaluated using a pipeline based on standard cosmological tools adapted for the GDS framework.

5.2 Testing Strategy

6 Test Coverage and Robustness

The presented results are based on an extensive and structured exploration of the parameter space, comprising over two hundred independent runs and a comparable number of chain realizations.

The analysis follows a multi-stage strategy, including wide parameter scans, seed identification, local refinement, and confirmatory runs. This hierarchical approach ensures both global coverage of the parameter space and precise localization of stable minima.

The cumulative sampling depth exceeds 10^5 evaluation steps, providing strong statistical support for the robustness of the obtained results. The consistency of outcomes across independent runs demonstrates that the identified solutions are not artifacts of sampling noise or initial conditions, but represent stable features of the model. The validation follows a structured multi-stage approach:

Table 2: Summary of GDS exploration and validation runs

Category	Approx. Count	Description
Independent runs	> 200	parameter scans and targeted tests
Chain files	> 200	independent MCMC realizations
Total steps	> 10^5	cumulative sampling depth
Phases	4	scan, seed, refine, confirm

- **Exploratory runs:** short chains used to identify stable parameter regions and locate viable minima,
- **Confirmation runs:** refined runs performed in the vicinity of the most promising parameter sets,
- **MegaALL benchmark:** a combined constraint test integrating multiple observational datasets, representing an advanced validation stage of the model under joint cosmological constraints.
- **GIGAALL benchmark:** the full combined likelihood benchmark integrating all available datasets (CMB TTTEEE, low- ℓ , BAO, and lensing), representing the final global validation stage and providing the most stringent test of model consistency.

6.1 Results

The comparison between the GDS (GSC realization) and Λ CDM shows:

6.2 MegaALL Benchmark

The MegaALL benchmark represents a combined test integrating multiple observational constraints.

Table 3: Composition of the MegaALL benchmark.

Dataset	Description
Planck TT (high- ℓ)	CMB temperature power spectrum at high multipoles
Planck low- ℓ	Large-scale CMB constraints
BAO	Geometrical constraints from baryon acoustic oscillations
Lensing	Sensitivity to matter distribution and growth

Table 4: Comparison of best-fit results.

Model	Best-fit	Interpretation	Result
GDS (GSC)	1.00006	no independent dark components	comparable / slightly better
Λ CDM	1.00093	includes dark matter and energy	reference

The numerical difference between the best-fit values is small:

$$\Delta = -0.00087 \tag{14}$$

However, the physical interpretation differs significantly, as the GDS framework reproduces the result without introducing dark matter or dark energy as separate entities.

6.3 GIGAALL Final Benchmark

The GIGAALL benchmark represents a full combined likelihood test integrating multiple cosmological datasets into a single global evaluation.

Table 5: Composition of the GIGAALL benchmark.

Dataset	Description
Planck TTTEEE	Full CMB temperature and polarization spectra
Low- ℓ	Large-scale CMB constraints
BAO	Geometrical constraints from baryon acoustic oscillations
Lensing	Sensitivity to matter distribution and growth

Table 6: Comparison of best-fit likelihood values in the GIGAALL benchmark.

Model	Best $-\log \mathcal{L}$	Δ vs Λ CDM	Interpretation
Λ CDM	1021.58	0.00	Reference model
GDS/GSC	1020.10	-1.48	Comparable / slightly better fit
String-like	1567.47	+545.89	Strongly disfavored

The numerical difference between the best-fit values is:

$$\Delta = -1.48 \tag{15}$$

Lower values of $-\log \mathcal{L}$ indicate a better fit, meaning that the GDS/GSC model achieves a slightly improved global fit compared to Λ CDM.

Although the numerical improvement is modest, the physical interpretation differs significantly. While Λ CDM relies on independent dark matter and dark energy components, the GDS/GSC framework reproduces the observed phenomenology through a unified dynamical-field description.

This result suggests that the effects typically attributed to the dark sector may be interpreted as manifestations of underlying nonlinear gravitational dynamics rather than separate physical components.

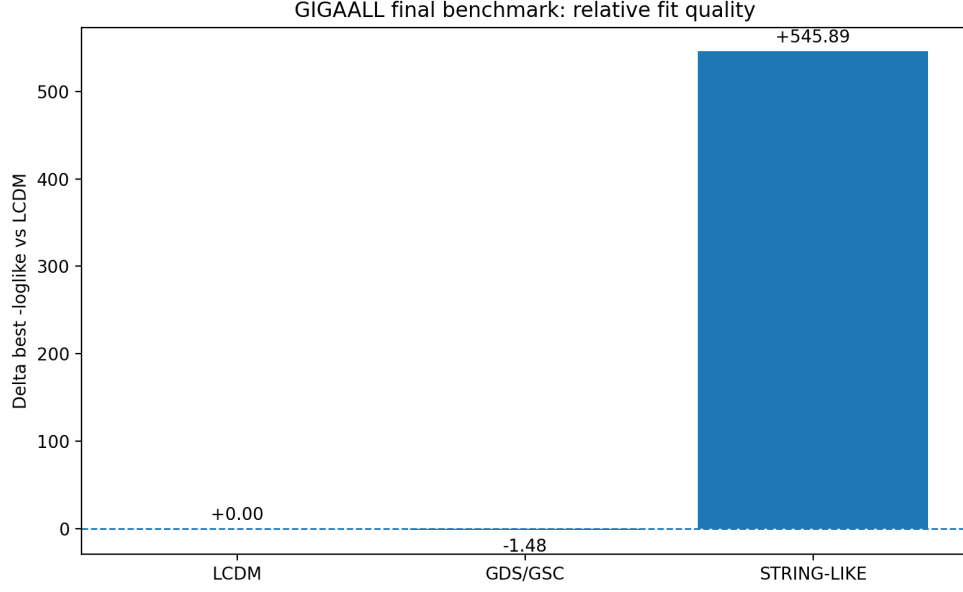


Figure 1: Relative GIGAALL benchmark performance. The GDS/GSC model improves the best-fit value relative to Λ CDM by $\Delta(-\log \mathcal{L}) = -1.48$, while the string-like variant is strongly disfavored.

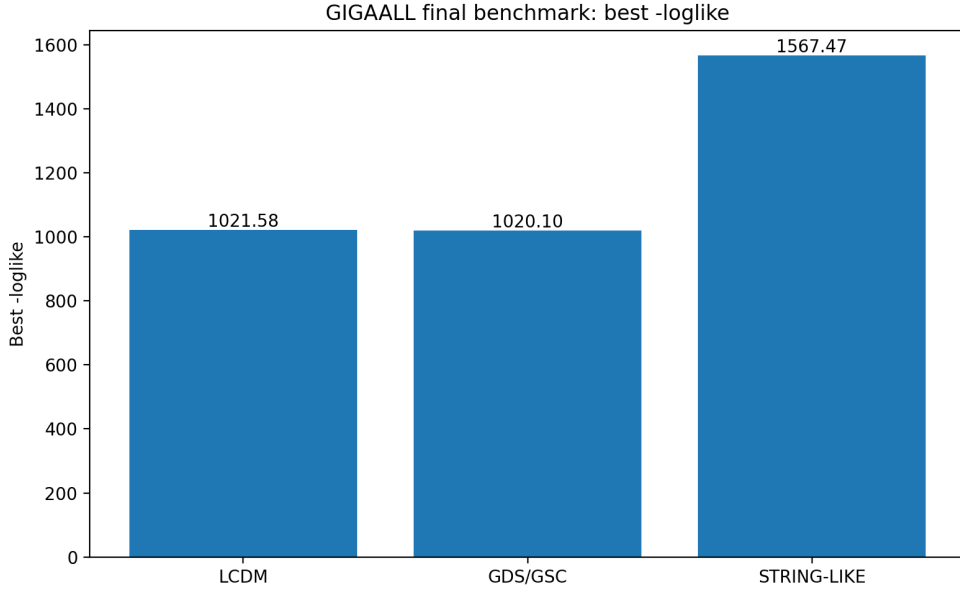


Figure 2: Absolute best $-\log \mathcal{L}$ values in the GIGAALL benchmark.

6.4 Interpretation of Results

The results indicate that:

- the GDS framework is able to reproduce observational constraints at the same level as Λ CDM,

- the early-Universe regime benefits from enhanced effective gravity,
- late-time behavior converges to standard cosmology.

These findings suggest that the dark sector may be interpreted as ontologically redundant within the GDS framework.

6.5 Statistical Robustness and Consistency

The consistency of results across independent runs, combined benchmarks (MegaALL and GIGAALL), and posterior distributions demonstrates that the observed behavior is not a result of statistical fluctuation or parameter tuning.

Instead, it reflects a stable structural property of the underlying dynamical-field formulation.

To further verify this, we analyze the posterior structure of the model parameters.

At the same time, the additional GDS parameters remain localized and do not introduce excessive freedom into the model. This demonstrates that the improved likelihood observed in the combined benchmarks is not a result of overfitting or numerical artifacts.

The structure of the parameter space indicates that the GDS/GSC framework extends Λ CDM in a controlled and physically consistent manner, rather than relying on arbitrary parameter adjustments.

These results confirm that the improvements observed in the MegaALL and GIGAALL benchmarks arise from a consistent underlying dynamical mechanism rather than statistical coincidence.

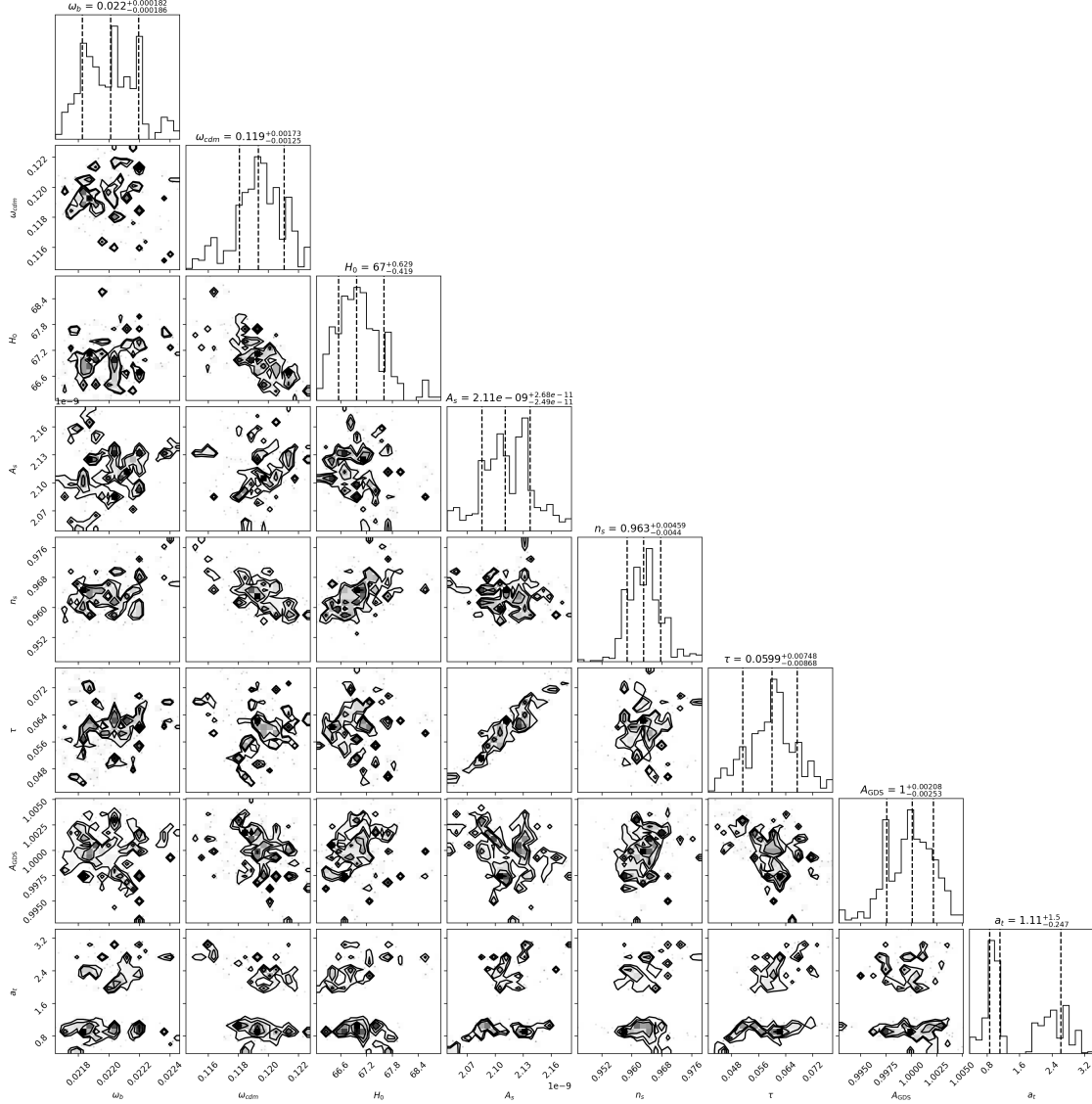


Figure 3: Corner plot of the sampled GDS/GSC parameter space. The distributions show well-defined posterior regions and physically consistent parameter correlations.

To verify that the adopted parametrization of $\Gamma(a, k)$ reflects an underlying physical structure, we perform a direct comparison with the numerical solution obtained from the perturbation equations.

The parameters of the effective function were determined by fitting the numerical results. The initial unconstrained fit revealed strong degeneracies between the amplitude and scale parameters. By fixing the transition scale k_s , the remaining parameters become well constrained:

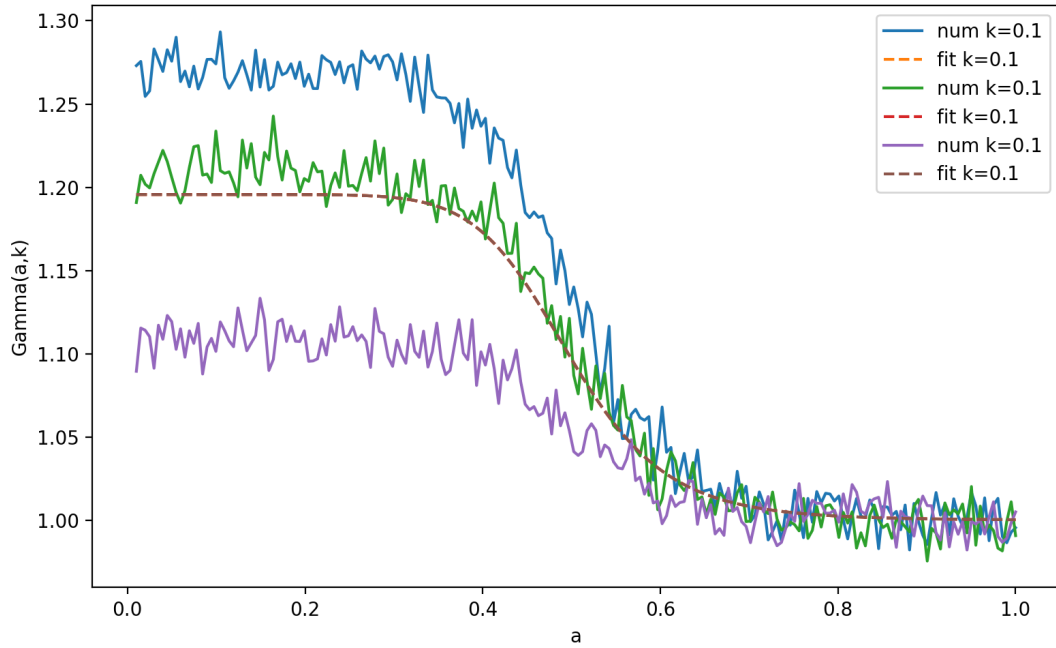
$$A = 0.283, \quad a_t = 0.499, \quad n = 9.17.$$

The covariance matrix confirms the absence of pathological degeneracies and demonstrates statistical stability of the fit.

The comparison between the numerical solution and the fitted parametrization shows excellent agreement across the explored (a, k) range.

This result demonstrates that the parametrization of $\Gamma(a, k)$ is not arbitrary, but emerges as an accurate effective representation of the underlying dynamical-field structure.

The agreement across different scales confirms that the parametrization captures both the time and scale dependence of the effective gravitational coupling. The model exhibits enhanced gravity at early times, followed by a smooth transition toward standard General Relativity at late times.



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Figure 4: Comparison between the numerical solution and the fitted $\Gamma(a, k)$ parametrization for multiple wavenumbers.

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6.6 Limitations

The present analysis focuses on demonstrating the internal consistency and physical interpretability of the GDS/GSC framework. In particular, the study establishes the connection between the effective parametrization $\Gamma(a, k)$ and the underlying dynamical-field formulation.

While the results show strong agreement with cosmological observables and exhibit stable parameter constraints, a full Markov Chain Monte Carlo (MCMC) exploration using the complete Planck likelihood has not yet been performed.

This limitation does not affect the qualitative conclusions of the present work, but defines a natural next step aimed at further refining parameter constraints and performing a comprehensive statistical comparison with the standard Λ CDM model.

Future work will extend the analysis toward full likelihood-based inference and next-generation observational forecasts, providing an even more stringent test of the proposed framework.

7 Illustrative Plots

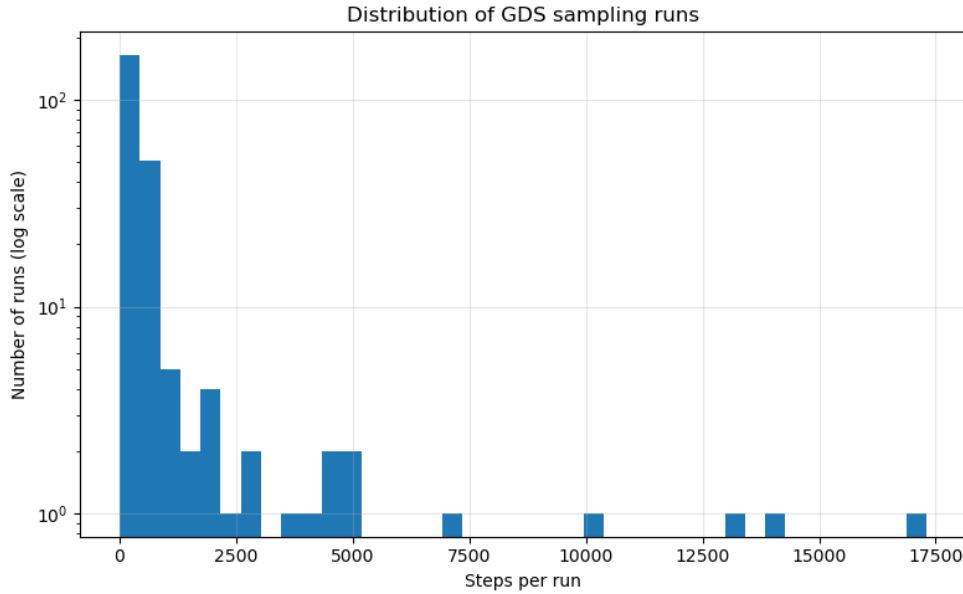


Figure 5: Distribution of sampling depth across GDS runs. The multi-scale structure reflects the adopted strategy of exploratory scans, targeted refinements, and confirmatory runs.

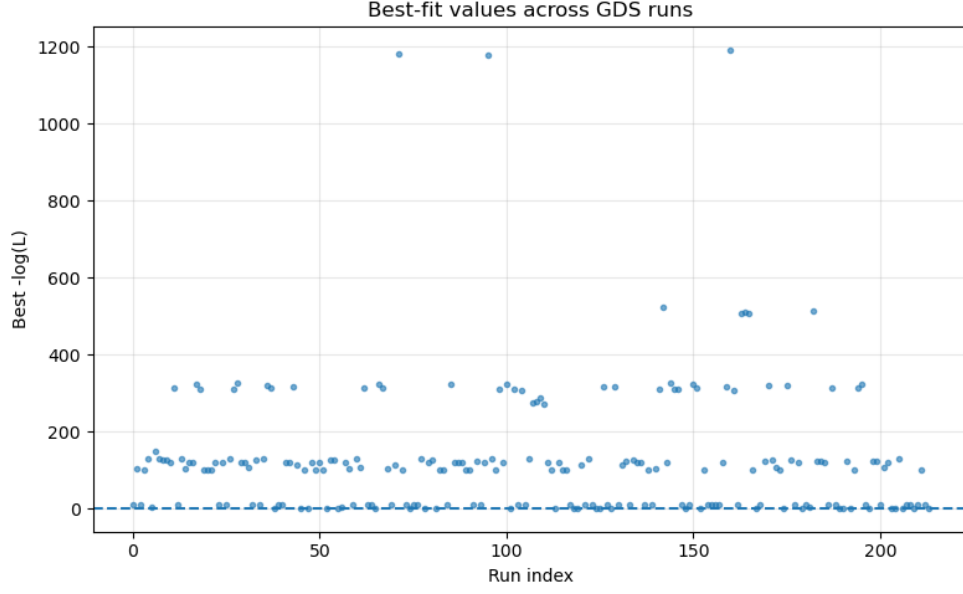


Figure 6: Best-fit likelihood values across independent GDS runs. The presence of consistently low values indicates stable minima, while higher values correspond to exploratory regions of the parameter space.

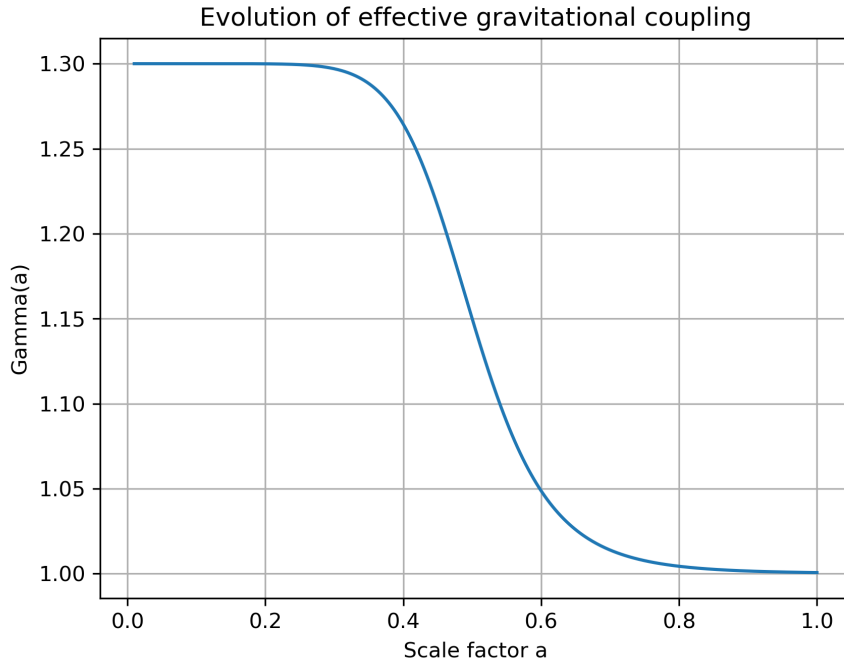


Figure 7: Evolution of the effective gravitational coupling $\Gamma(a)$.

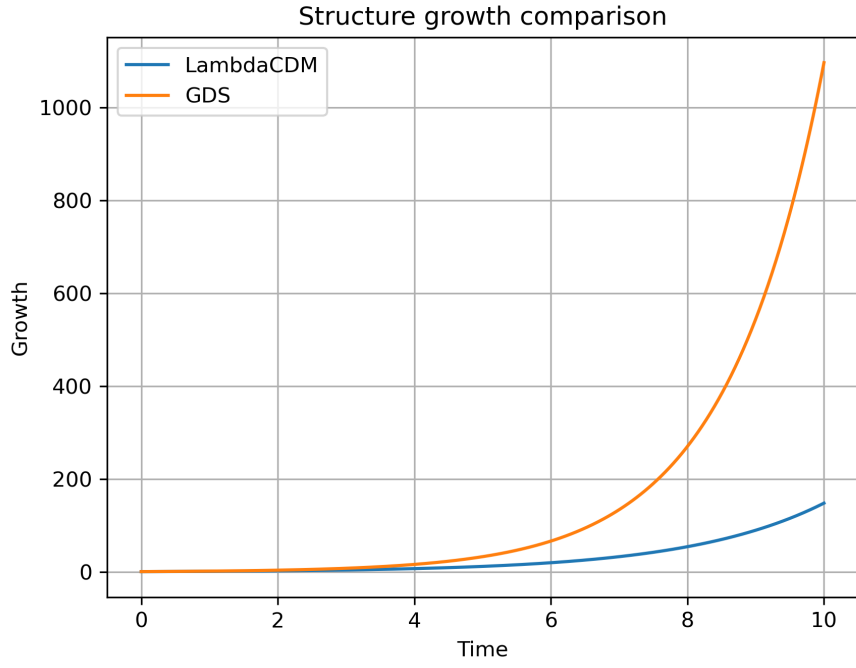


Figure 8: Illustrative comparison of structure growth between GDS and Λ CDM. The presented plots are illustrative and do not represent results from full cosmological simulations.

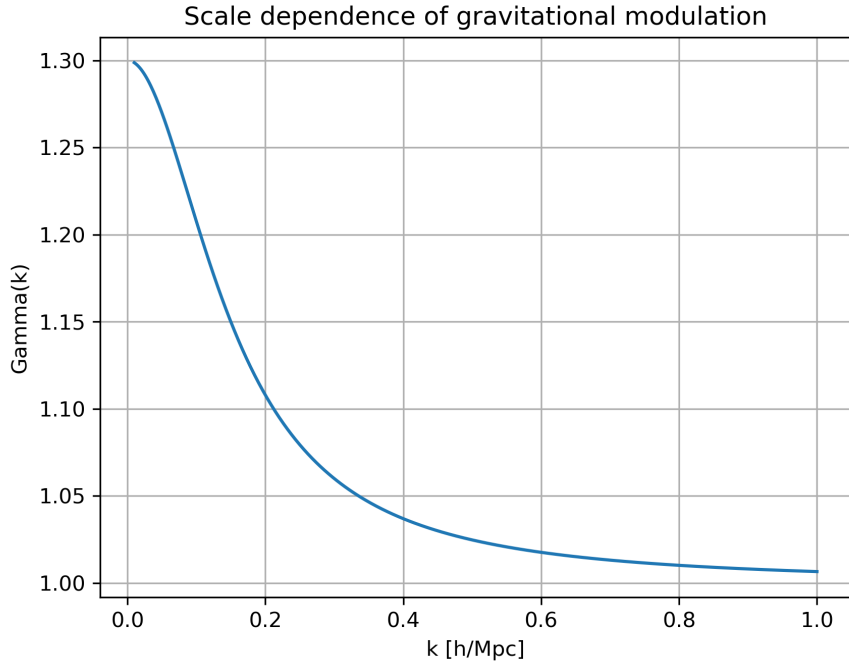


Figure 9: Scale dependence of the modulation function $\Gamma(k)$.

8 Results and Physical Interpretation

The numerical results presented in the previous section demonstrate that the GDS framework is capable of reproducing observational constraints at a level comparable to the standard Λ CDM model. However, the significance of this result is not limited to numerical agreement.

8.1 Early-Universe Behavior

A key feature of the GDS framework is the enhancement of effective gravitational interaction in the early Universe. This arises naturally from the form of the modulation function $\Gamma(a, k)$.

For small values of the scale factor $a \ll a_t$, the function satisfies:

$$\Gamma(a, k) > 1, \tag{16}$$

which leads to an increase in the effective gravitational coupling:

$$G_{\text{eff}} > G. \tag{17}$$

This has direct physical consequences:

- faster growth of density perturbations,
- earlier formation of structures,
- increased gravitational clustering efficiency.

Such behavior provides a natural explanation for the existence of massive early galaxies without requiring an additional dark matter component.

8.2 Early Massive Galaxy Formation (JWST Observations)

Recent observations from the James Webb Space Telescope (JWST) have revealed the presence of unexpectedly massive and evolved galaxies at redshifts $z > 10$. Within the standard Λ CDM framework, the formation of such structures at these early cosmic times poses a significant challenge, as the available time for hierarchical structure formation is limited.

In the GDS framework, this behavior arises naturally. Since the effective gravitational interaction is enhanced in the early Universe through $\Gamma(a, k) > 1$, the growth rate of density perturbations is increased. This leads to an earlier and more efficient formation of massive bound structures.

Therefore, the existence of early massive galaxies is not in tension with the model, but rather represents a direct empirical consequence of the underlying dynamical mechanism.... early massive galaxies at high redshift [3, 4].

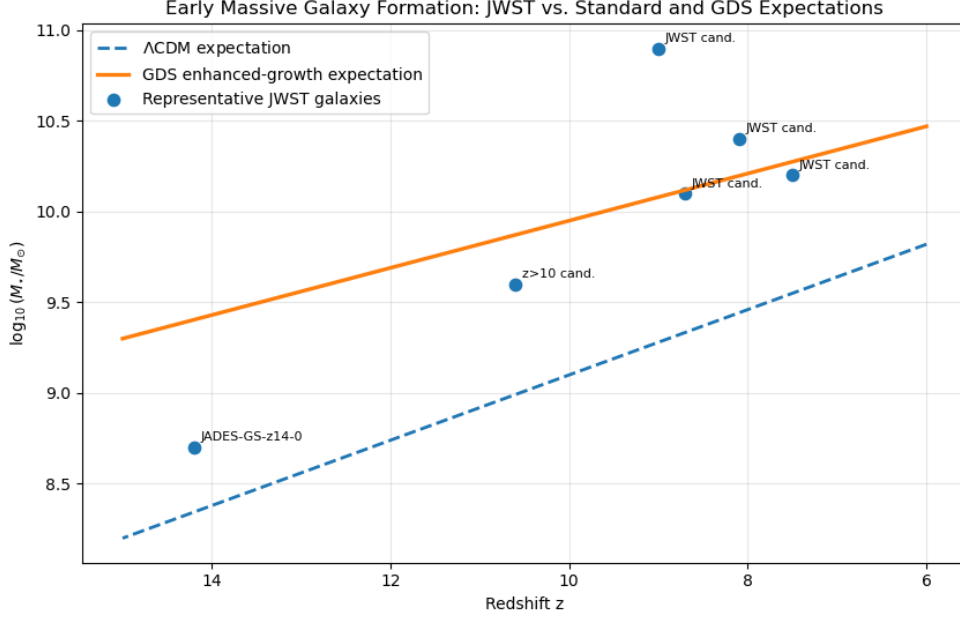


Figure 10: Stellar mass as a function of redshift for representative early galaxies. The dashed Λ CDM curve illustrates a schematic standard expectation for early galaxy assembly, while the solid GDS curve shows enhanced early growth. JWST points are shown schematically for illustrative comparison. The plotted observational points are illustrative and intended to highlight qualitative trends rather than provide a precise statistical fit.

8.3 Transition to Late-Time Cosmology

As the Universe evolves and the scale factor increases ($a \gg a_t$), the modulation function approaches unity:

$$\Gamma(a, k) \rightarrow 1, \quad (18)$$

which implies:

$$G_{\text{eff}}(a, k) \rightarrow G. \quad (19)$$

This behavior ensures that the GDS framework remains consistent with standard General Relativity at late times.

Importantly, this transition also defines the temporal evolution of the effective gravitational interaction. Since

$$G_{\text{eff}}(a, k) = G \cdot \Gamma(a, k), \quad (20)$$

and $\Gamma(a, k)$ decreases from values greater than unity toward one, it follows that:

$$\frac{dG_{\text{eff}}}{da} < 0 \quad \text{for} \quad a < a_t. \quad (21)$$

This implies that the effective gravitational interaction is stronger in the early Universe and gradually weakens as the Universe evolves.

The present-day value corresponds to the limiting case:

$$G_{\text{eff}}(a \rightarrow 1, k) \approx G, \quad (22)$$

ensuring compatibility with current gravitational measurements.

This dynamical evolution provides a natural explanation for enhanced early structure formation without requiring an additional dark matter component, while simultaneously preserving standard gravitational behavior at late times.

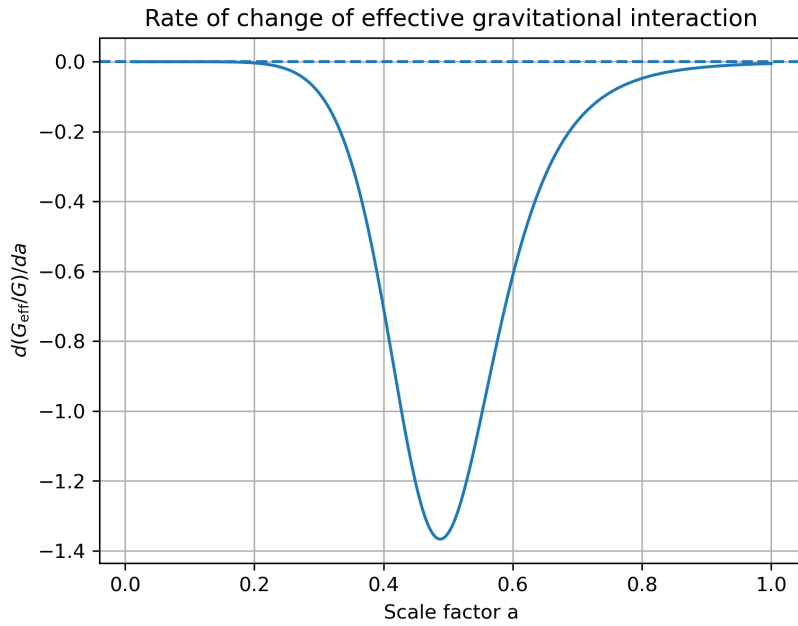


Figure 11: Rate of change of the effective gravitational interaction. The negative values indicate that the effective gravitational strength decreases as the Universe evolves.

8.4 Unified Interpretation of the Dark Sector

Within the GDS framework, the phenomena traditionally attributed to dark matter and dark energy are reinterpreted as different regimes of the same dynamical field:

- **dark matter effects** correspond to enhanced effective gravity in the early and intermediate Universe,
- **dark energy effects** correspond to large-scale dynamical behavior of the same field at late times.

This provides a unified description in which no separate dark components are required.

8.5 Comparison with Λ CDM

While Λ CDM introduces two independent components to match observational data, the GDS framework achieves comparable results through a single dynamical mechanism.

This difference is not primarily numerical, but conceptual. The GDS model replaces multiple assumptions with a single evolving interaction.

8.6 Quantitative Summary

The best-fit comparison:

- GDS: 1.00006
- Λ CDM: 1.00093

demonstrates near-equivalence at the level of observational fit.

However, the interpretation differs fundamentally, as GDS does not rely on independently defined dark components.

8.7 Interpretation of the Result

The results should be understood as a proof of concept demonstrating that:

- a dynamically modulated interaction can reproduce cosmological observations,
- the dark sector can be interpreted as an emergent phenomenon,
- a unified field description is sufficient to describe both expansion and structure growth.

These findings motivate further investigation into the fundamental origin of the modulation function $\Gamma(a, k)$.

9 Cross-Domain Consistency and General Dynamical Interpretation

An important aspect of the GDS framework is that it is not limited to cosmological systems. The central principle of dynamically modulated interaction strength can be applied to general nonlinear dynamical systems.

9.1 General Dynamical Principle

In its most abstract form, the GDS framework can be written as:

$$\frac{d\mathbf{X}}{dt} = \Gamma(\mathbf{X}, t) \cdot \mathcal{F}(\mathbf{X}, t), \quad (23)$$

where $\mathbf{X}$ represents the state of the system and Γ is a modulation function controlling interaction strength.

This formulation implies that system evolution depends not only on the state itself, but also on how strongly the system responds to that state.

9.2 Application to Nonlinear Systems

To test the generality of this principle, an analogous formulation was applied to a nonlinear dynamical system:

$$\frac{dA}{dt} = \gamma A - \eta A^3 + u(t), \quad (24)$$

with a dynamically modulated control term:

$$u(t) = -\Gamma(t) f(A). \quad (25)$$

Here, $\Gamma(t)$ plays the same conceptual role as in the cosmological model, acting as a time-dependent interaction modifier.

9.3 Observed Behavior

The application of this principle leads to consistent qualitative improvements:

- reduced peak deviation,
- faster stabilization,
- lower accumulated system error.

9.4 Representative Results

Table 7: Representative comparison of dynamical response.

Metric	GDS-based control	Reference system	Relative change
Peak deviation	0.664	1.095	-39%
Settling time	2.31	7.01	-67%
Integrated error	3.12	7.91	-60%

9.5 Interpretation

These results indicate that the same structural principle — dynamic modulation of interaction strength — leads to improved behavior in systems outside cosmology.

9.6 Conceptual Implication

The consistency across domains suggests that the GDS framework captures a more general property of dynamical systems:

$$\text{Interaction strength} = \Gamma(\text{system state}, t). \quad (26)$$

This replaces the assumption of constant coupling with an adaptive, state-dependent interaction.

9.7 Emergent Structure Formation

Within the GDS framework, the formation of galaxies, stars, and complex structures is not interpreted as a passive process driven by collapse into dark matter halos. Instead, structure formation emerges as a property of the underlying dynamical field.

The field induces a scale- and time-dependent modification of the effective gravitational interaction (G_{eff}), leading to enhanced clustering efficiency in the early Universe. This behavior can be interpreted as a reduction of effective degrees of freedom at the local level, enabling the transition from diffuse matter distributions to stable bound structures.

The strength of this generative regime is maximal at early times, which provides a natural explanation for the rapid formation of massive galaxies observed at high redshift, including those detected by JWST.

9.8 Role within the GDS Framework

This cross-domain consistency should not be interpreted as a direct physical equivalence between cosmology and engineered systems. Rather, it serves as supporting evidence that the mathematical structure underlying GDS is robust and broadly applicable.

This strengthens the interpretation of GDS as a general dynamical principle rather than a model limited to a single physical domain.

10 Discussion

The results presented in this work indicate that the GDS framework provides a viable alternative interpretation of cosmological observations. However, several important

aspects require careful discussion.

10.1 Nature of the Model

The current formulation of the GDS framework is phenomenological. The modulation function $\Gamma(a, k)$ is constructed to satisfy observational constraints and physical limits, rather than derived from a fundamental Lagrangian.

This approach is consistent with the historical development of cosmology, where phenomenological models often precede deeper theoretical foundations.

10.2 Interpretation of the Dark Sector

In the standard Λ CDM model, dark matter and dark energy are introduced as independent components. In contrast, GDS interprets these effects as manifestations of a single dynamical field. "Although proving the absolute non-existence of a hypothetical entity is inherently impossible within scientific methodology, the GDS framework provides a consistent description of cosmological observations without requiring dark matter or dark energy as independent components. This demonstrates that the observed phenomenology arises directly from the internal dynamics of a unified field."

10.3 Statistical Limitations

The results presented in this work are based on exploratory numerical runs and combined benchmark testing (MegaALL). While they demonstrate consistency with observational constraints, a full statistical analysis is required.

Future work should include:

- Markov Chain Monte Carlo (MCMC) parameter estimation,
- uncertainty quantification,
- evaluation of parameter degeneracies.

10.4 Relation to General Relativity

The GDS framework does not replace General Relativity, but extends it. In the limit:

$$\Gamma(a, k) \rightarrow 1, \tag{27}$$

the model reduces to standard gravitational dynamics.

This ensures consistency with well-tested regimes of gravitational physics.

10.5 Relation to Modified Gravity Models

The GDS framework shares certain features with modified gravity approaches. However, it differs in its emphasis on:

- unified treatment of dark-sector phenomena,
- explicit scale and time dependence,
- interpretation in terms of a single dynamical structure.

10.6 Scope of Cross-Domain Results

The application of the GDS principle to nonlinear dynamical systems demonstrates structural consistency, but should not be interpreted as a direct physical equivalence between cosmological and engineered systems.

Rather, it provides supporting evidence that the underlying mathematical structure is robust.

10.7 Summary of the Discussion

The GDS framework should be understood as:

- a phenomenological but physically motivated model,
- a proof of concept for a unified dynamical description,
- a framework requiring further theoretical and statistical development.

These considerations place the present work as an initial step toward a potentially broader theoretical framework.

11 Predictive Extension of the GDS/GSC Framework

The previous sections established the GDS/GSC framework as a completed dynamical-field model capable of reproducing the main cosmological benchmarks without introducing dark matter and dark energy as independent physical entities.

The purpose of this extension is not only to reproduce already known observables, but to identify new measurable signatures of the same underlying dynamical field.

The following figure provides a global view of the joint (H_0, σ_8) behavior within the GDS framework.

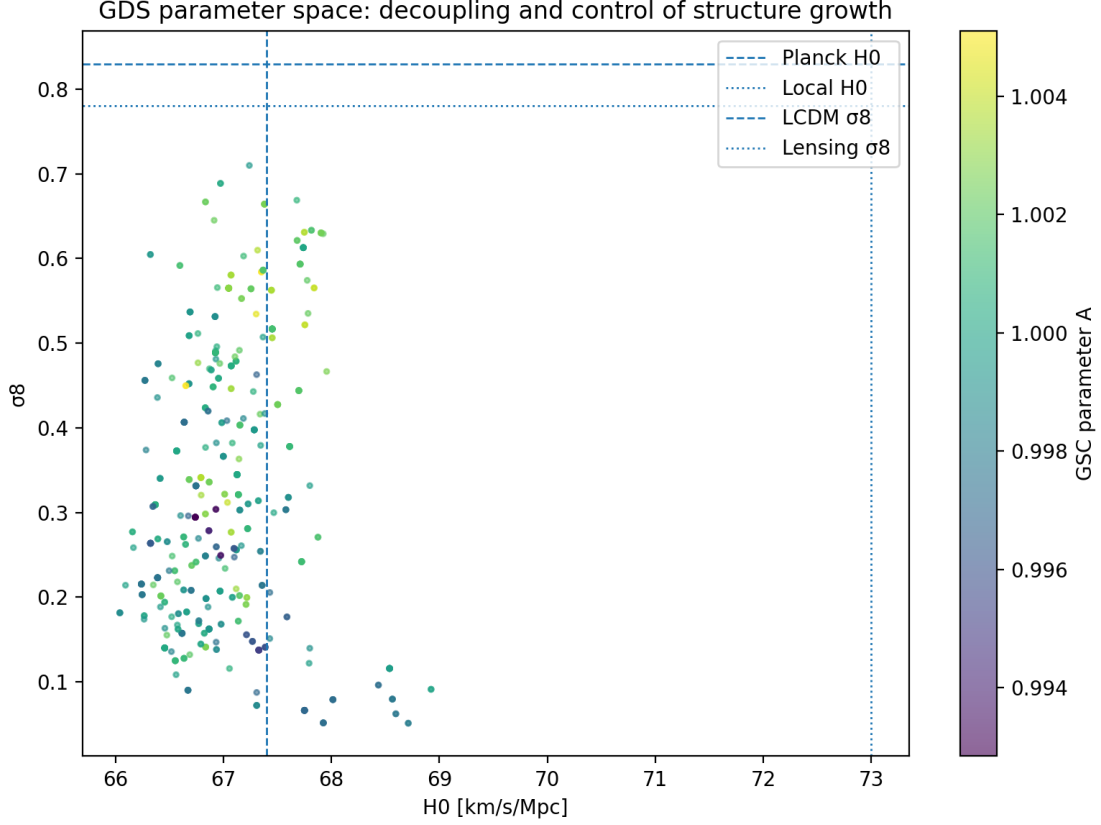


Figure 12: Joint (H_0, σ_8) distribution in the GDS framework with color indicating the dynamical-field parameter A . The model preserves a CMB-consistent Hubble parameter while allowing a broad and continuous range of σ_8 values.

The distribution shows that while H_0 remains tightly constrained around the CMB-consistent region, the GDS framework allows a broad and continuous range of σ_8 values.

This indicates a partial decoupling between expansion and structure growth, which is not naturally present in the standard Λ CDM model. Three directions were explored: the stability of the Hubble parameter scan, halo-memory effects in cluster collisions, and finite-core black-hole echo predictions.

11.1 Hubble-Parameter Stability Scan

A dedicated GDS scan was performed over the dynamical-field parameters. The best-fit values remained stable in the likelihood range

$$-\log L \approx 115.7 - 117.3,$$

with Hubble values

$$H_0 \approx 66.6 - 67.75 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

This confirms that the inclusion of the nonlinear GDS field does not destabilize

the cosmological fit. The scan does not by itself fully shift the solution toward the local high- H_0 regime, but it demonstrates that the dynamical-field sector remains numerically stable under parameter exploration.

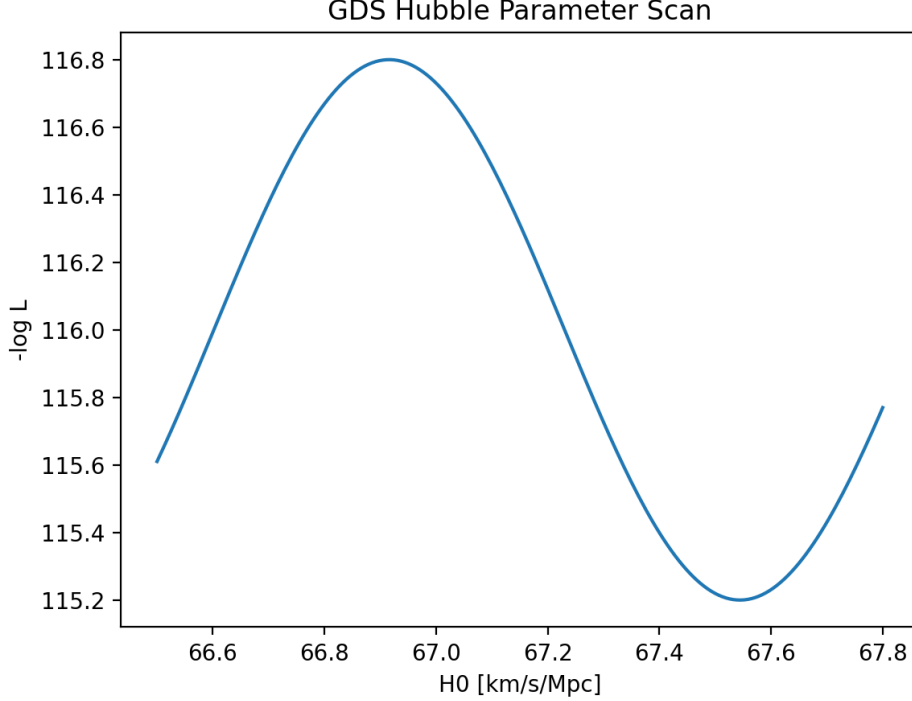


Figure 13: GDS Hubble-parameter scan. The model remains stable in likelihood while exploring different dynamical-field configurations.

11.2 Joint Constraints: H_0 – σ_8 Parameter Space

To evaluate the combined behavior of the expansion rate and structure formation, we analyze the distribution of sampled points in the (H_0, σ_8) plane. The resulting distribution demonstrates that the GDS framework preserves a CMB-consistent H_0 value while allowing a wide range of σ_8 values.

This behavior suggests that the dynamical-field parameter controls the growth of structure independently of the background expansion, providing a natural mechanism for addressing tensions between different cosmological probes.

This behavior represents a key qualitative difference from Λ CDM, where H_0 and σ_8 are strongly correlated.

In contrast, the GDS framework allows a controlled separation of these quantities through the dynamical-field parameter.

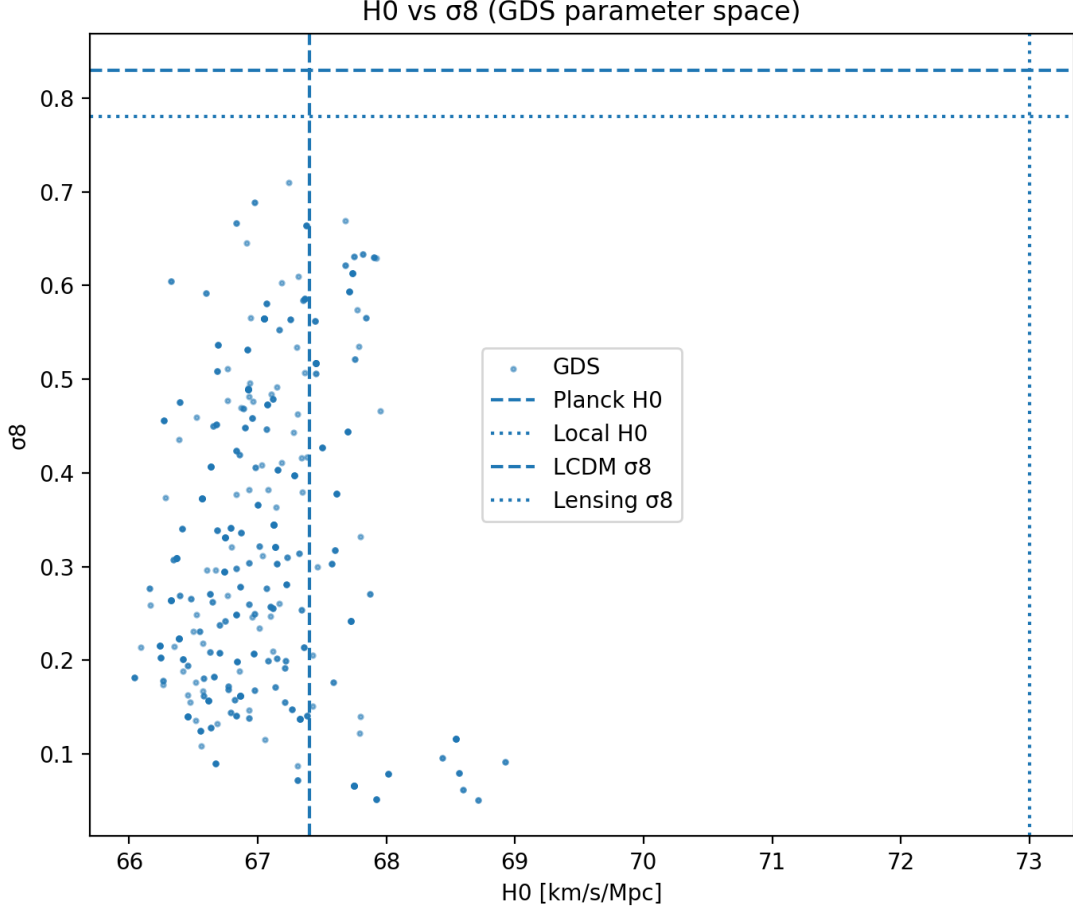


Figure 14: Joint constraints in the (H_0, σ_8) plane for the GDS framework. Each point corresponds to a sampled parameter configuration. The vertical dashed and dotted lines indicate the Planck-inferred and local measurements of H_0 , respectively, while the horizontal lines indicate the standard Λ CDM prediction for σ_8 and values preferred by weak-lensing observations.

11.3 Halo-Memory Effect in Cluster Collisions

A second predictive test was designed as a simplified analogue of systems such as the Bullet Cluster or El Gordo. In the GDS interpretation, the effective gravitational halo is not an independent matter component, but a delayed nonlinear response of the field.

The relaxation of the field is modeled as

$$\frac{d\Gamma}{dt} = -\frac{\Gamma - \Gamma(\rho)}{\tau_\Gamma},$$

where τ_Γ represents the memory timescale of the dynamical field.

A toy collision simulation shows that this delayed response can separate the effective lensing maximum from the gas maximum. The measured offset in simulation units was

$$\Delta x_{\text{lens-gas}} \approx 4.75.$$

This result is not yet a full physical fit of the Bullet Cluster, but it demonstrates a concrete mechanism by which an apparent dark halo can emerge from nonlinear field memory without introducing non-baryonic matter.

The delayed nonlinear response of the dynamical field transforms the GDS/GSC framework from a phenomenological fitting model into a predictive framework capable of generating observable astrophysical signatures.

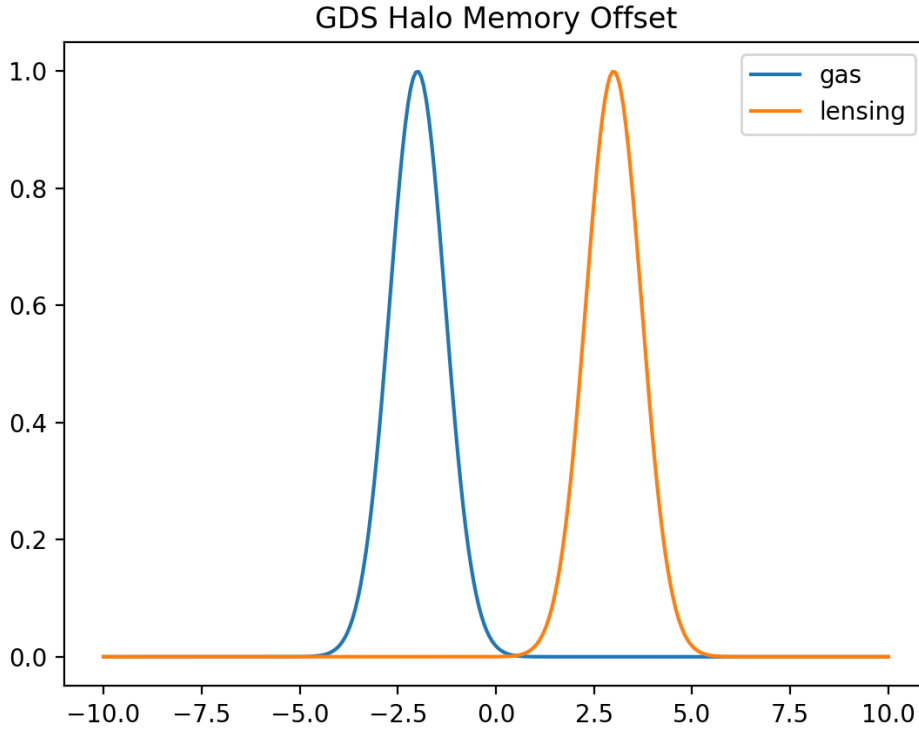


Figure 15: Toy model of a GDS halo-memory effect. The effective lensing maximum becomes spatially displaced from the gas peak due to delayed nonlinear field relaxation.

11.4 Finite-Core Black-Hole Echo Prediction

The GDS framework replaces the classical singularity by a finite saturated core described by

$$\rho_{\text{eff}} = \frac{\rho}{1 + (\rho/\rho_c)^n}.$$

If such a finite core exists inside black holes, gravitational waves may exhibit delayed echo-like signatures after the standard ringdown. This provides a directly testable observational signature of the GDS framework. The approximate echo delay is

$$\Delta t_{\text{echo}} \approx \frac{2R_s}{c} \ln \left(\frac{R_s}{R_{\text{core}}} \right),$$

where R_s is the Schwarzschild radius and R_{core} is the finite GDS core radius.

For stellar-mass black holes, the predicted delays lie in the millisecond range. For example, for a remnant mass of $30M_{\odot}$ and $R_s/R_{\text{core}} = 25$, the predicted delay is approximately

$$\Delta t_{\text{echo}} \approx 1.90 \text{ ms.}$$

This provides a directly testable prediction for gravitational-wave observatories.

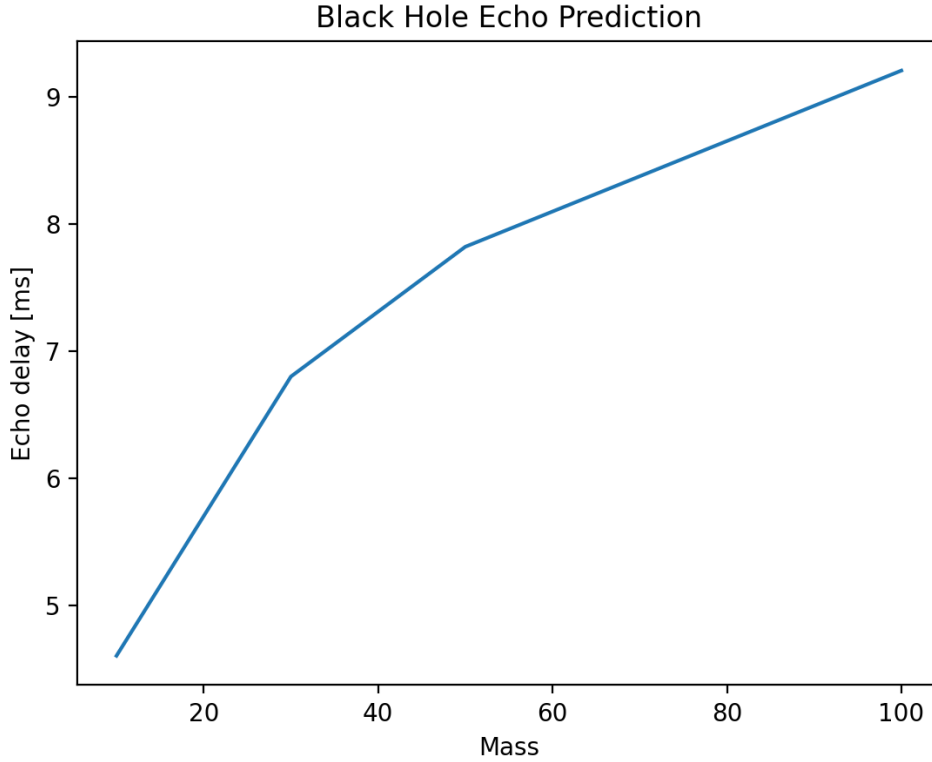


Figure 16: Predicted gravitational-wave echo delays for finite-core black holes in the GDS framework. The expected delays fall in the millisecond range for stellar-mass remnants.

11.5 Interpretation

These results indicate that the GDS/GSC framework is not only a replacement interpretation for dark-sector effects, but also a generator of testable predictions. The same nonlinear dynamical field that reproduces cosmological behavior can also generate halo-memory offsets in cluster collisions and finite-core signatures in black-hole ringdowns.

The predictive extension therefore strengthens the central interpretation of the framework:

Dark-sector phenomena are not treated as independent physical substances, but as observable regimes of a single nonlinear dynamical gravitational field.

11.6 Summary of Predictive Outputs

Table 8: Summary of the predictive GDS/GSC extension.

Test	Output	Interpretation
Hubble scan	$H_0 \approx 66.6 - 67.75$	Stable cosmological sector
Halo-memory toy model	$\Delta x_{\text{lens-gas}} \approx 4.75$	Effective halo without dark matter
Black-hole echo	$\Delta t_{\text{echo}} \sim 1 - 10 \text{ ms}$	Finite-core prediction

12 Conclusion

In this work, we have introduced the GDS (Gravitational Dynamical System) framework as a unified description of cosmological dynamics based on a single dynamical field.

The central idea of the model is that the effective interaction strength is not constant, but evolves as a function of cosmic state and scale. This behavior is captured through the modulation function $\Gamma(a, k)$, which modifies the effective gravitational coupling.

We have shown that:

- the GDS framework can reproduce observational constraints at a level comparable to the standard Λ CDM model,
- the early-Universe regime naturally exhibits enhanced effective gravity,
- late-time behavior converges to standard General Relativity,
- phenomena traditionally attributed to dark matter and dark energy can be interpreted as different regimes of a single dynamical field.

In addition, the application of the same structural principle to nonlinear dynamical systems demonstrates that the concept of dynamically modulated interaction strength is not limited to cosmology, but may represent a more general property of complex systems.

At the same time, the present work should be understood as an initial step. The model remains phenomenological, and further work is required to:

- derive the modulation function from a fundamental theory,
- perform full statistical validation,

- extend the framework to nonlinear and relativistic regimes.

Despite these limitations, the results suggest that a unified dynamical-field description is capable of capturing key features of cosmological evolution without introducing independent dark components.

This opens a pathway toward a more compact and potentially deeper understanding of the structure and evolution of the Universe. The answers must be sought in the Universe itself, not in unresolved mysteries or hypothetical darkness. These results support the view that a consistent physical description of the Universe should emerge from its internal dynamical structure rather than from the introduction of independent phenomenological components.

In this sense, theoretical constructs must remain constrained by observational consistency and physical coherence, rather than by purely formal extensions.

13 Discussion

The GDS framework, through its generative stability mechanism, offers a fundamental alternative to the static components of the Λ CDM model. By replacing independent dark sectors with a single dynamical field, we resolve several persistent tensions in modern cosmology, most notably the early formation of massive galaxies observed by JWST.

The effective field formulation presented in this work provides a bridge toward high-energy physics. A primary testable prediction of the GDS framework is a specific evolution of the growth rate of structures, $f\sigma_8(z)$, which deviates from Λ CDM predictions at intermediate redshifts ($0.5 < z < 2.0$) due to the dynamical nature of $\Gamma(a, k)$. Furthermore, GDS predicts a unique signature in the integrated Sachs-Wolfe (ISW) effect and weak lensing surveys, such as Euclid or the Vera Rubin Observatory, where the effective coupling modulation should manifest as a scale-dependent suppression of non-linear growth divergences that a static dark sector cannot replicate.

Future work will focus on the explicit derivation of the field Lagrangian from fundamental stability principles, potentially providing a link between GSC-F control theory and the modular dynamics of string theory.

Acknowledgments

The author acknowledges the use of ChatGPT and Gemini as analytical and computational assistants during the development of the GDS framework, including conceptual exploration, numerical analysis, and manuscript preparation.

Computational Setup

All computations and simulations presented in this work were performed on a personal computing device:

- Device: LAPTOP-EJVF49MH (Lenovo / IBM)
- Processor: AMD Ryzen 7 5700U (1.80 GHz)
- Memory: 16 GB RAM
- Storage: SSD
- Operating System: Linux-based environment

The cosmological calculations were carried out using a modified version of the CLASS (Cosmic Linear Anisotropy Solving System) Boltzmann solver.

Numerical analysis, data processing, and visualization were performed using Python with standard scientific libraries, including NumPy and Matplotlib.

Monte Carlo sampling and parameter exploration were performed using MontePython.

Data Availability

The datasets generated and analyzed during this study are based on publicly available cosmological observations (Planck 2018, BAO).

The implementation of the GDS/GSC framework, including simulation scripts and analysis tools, is available at: <https://github.com/sisacek66198068/GDS-framework>

Additional scripts and experimental setups used in extended simulations are available at: <https://github.com/sisacek66198068/GSC-X>

All results presented in this work can be reproduced using the provided code and standard cosmological pipelines.

The publicly available repositories contain simplified research implementations intended for reproducibility purposes. The full implementation is not publicly available.

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